TECHNICAL REVIEW AND PROPOSED EVALUATION PROTOCOL

Evaluating Long-Form Speech Transcription

Segmentation, punctuation, timestamps, and reproducible assessment

Abstract

Long-form transcription is often judged by the accuracy of its words, yet a usable transcript also depends on where audio is divided, how segments are joined, whether punctuation is supplied, and how faithfully timestamps locate speech. This review explains how these components interact and proposes a practical protocol for evaluating them. It distinguishes fixed-duration segmentation from pause-aware segmentation, discusses the limits of comparing compressed formats, and describes separate measures for recognition, punctuation, timing, and operational reliability. Particular attention is given to preserving reference text, recording model and preprocessing settings, and testing the same speech under paired conditions. The paper is a technical review and proposed methodology; it reports no new dataset, experiments, or product performance claims.

Keywords: automatic speech recognition; long-form audio; segmentation; punctuation restoration; evaluation; reproducibility

1. Introduction

Transcribing a short, clear clip is a different engineering problem from delivering a reliable transcript of a meeting or interview. In a long recording, speakers pause, interrupt one another, change topics, and move closer to or farther from the microphone. A production service must also handle files, processing limits, retries, and a transcript that people can actually read.

Recent speech recognition systems have made broad improvements in robustness, but their output is not immune to variation in the audio or in the processing pipeline. Whisper, for example, was trained using large-scale weak supervision across a range of tasks and conditions [1]. That general approach does not imply that every implementation using a Whisper-family model will behave identically: wrappers may choose different decoding settings, voice activity detection, timestamps, and post-processing.

This distinction matters when an output contains correct words but little or no punctuation. It may be tempting to attribute the problem to the duration of the file, a particular cut point, or the model itself. Any of these may be relevant in a given system, but a single observation cannot isolate the cause. A useful investigation holds most of the pipeline constant and changes one condition at a time.

This article offers a framework for doing that. It is not an empirical comparison of transcription services, and it makes no claim that a particular segment length or audio format solves punctuation problems.

2. From recording to readable transcript

A typical workflow has several stages: media decoding, audio normalization, optional voice activity detection (VAD), segmentation, speech recognition, timestamp alignment, segment merging, and presentation or export. Some systems combine stages within one model; others use separate services. The distinction is important because punctuation may be generated by the recognizer, added by a second model, altered during merging, or lost during export.

For diagnosis, retain a copy of the output at each stage where this is feasible and authorized. Compare the raw model response with the stored transcript, the displayed text, and the exported file. If punctuation is present in the model response but absent in the interface, an additional punctuation model will not address the actual failure. Conversely, if punctuation is already absent in the model response, a display-layer fix alone cannot restore it.

VAD deserves particular attention. Removing silence may reduce the amount of audio sent to a recognizer, but it can also change the timing or context around speech. A VAD setting is therefore part of the experimental condition, not merely an implementation detail. Record whether it was enabled, its version and settings, and how timestamps were mapped back to the original file.

3. Choosing and testing segment boundaries

3.1 Fixed intervals and speech-aware boundaries

Fixed intervals are easy to reproduce: for example, a system might process successive 10-minute portions. Their weakness is that a boundary can fall in the middle of a word, sentence, or speaker turn. Pause-aware segmentation moves a boundary toward a suitable pause, subject to the service's length and size limits. This can preserve more linguistic context, but a pause is not necessarily the end of a sentence, and a noisy recording may contain pauses that VAD misses.

A practical policy is to set a maximum duration and search for an acceptable pause within a bounded window before that limit. If no pause is found, the system can fall back to a fixed boundary. The chosen window, VAD thresholds, and fallback rule should be recorded. A small overlap between adjacent segments may reduce lost speech at the boundary, but it introduces a second problem: duplicate words must be reconciled when the transcript is merged.

Neither a fixed 20-minute limit nor any other single duration can guarantee punctuation. Shorter segments may help avoid a service limit or reduce the scope of a failure, but they also create more boundaries and more requests. The appropriate setting depends on the model, file constraints, workload, and measured results.

3.2 A controlled boundary experiment

To test whether a boundary contributes to a problem, select recordings long enough to support all planned conditions. For the same source recording, compare a full-file run where available, fixed-length segments, and pause-aware segments at the same nominal length. Repeat selected runs to detect nondeterministic behavior. Keep model version, language setting, prompt, decoding options, normalization, and output processing unchanged.

Inspect a boundary window as well as the whole transcript. For example, examine a manually chosen region around each cut for missing or repeated words and punctuation changes. Report the window definition in advance and do not silently discard failed or empty segments. A single short

clip cannot establish a long-form result; the same comparison should be repeated across multiple recordings and speakers.

4. Audio formats are not interchangeable test conditions

WAV, FLAC, MP3, and M4A are useful labels for users, but a filename extension does not fully describe the audio. WAV and M4A are containers that can carry different codecs; FLAC is a lossless codec, while common MP3 and AAC configurations use lossy compression. Sample rate, channel count, bitrate, and encoder settings also matter. A fair comparison records the actual codec and encoding parameters, not only the extension.

Generate each format from the same authorized source file. Avoid converting MP3 to AAC and calling the result an independent M4A condition, because that adds another lossy generation. Instead, encode both from the same master. Preserve the original unchanged, store a checksum for each derivative, and document the command and software version. When comparing identical speech in different formats, use the same reference transcript and check that decoded duration and channel handling are consistent.

A format comparison should not be interpreted as a comparison of recording quality in the wild. A clean WAV and a noisy phone recording differ in microphone, environment, and production history as well as format. Those factors require their own test design.

5. Designing a representative evaluation set

LibriSpeech provides a documented English speech corpus and a useful starting point for reproducible recognition tests [2]. Its read-audiobook material does not, by itself, represent spontaneous meetings or interviews. A benchmark intended for those uses should include authorized conversational audio, varied microphones and environments, different speaking rates, and recordings of the lengths the service is expected to process.

Publicly available short clips can be concatenated for a pipeline stress test, provided the applicable terms allow the intended use. Such a file tests throughput and handling of long inputs, but it should not be described as a naturally occurring long conversation. Speaker transitions, pauses, and context are artifacts of the construction. Keep synthetic or concatenated tests separate from genuine continuous recordings in any report.

Representation matters as well as duration. Koenecke and colleagues documented disparities across speaker groups in several commercial ASR systems [3]. Their findings do not establish the performance of a different system, but they illustrate why a single aggregate score can obscure uneven performance. Choose sample categories that fit the research question, report their sizes, and avoid making demographic claims from small or unrepresentative groups.

Reference transcripts should be created or checked by qualified human reviewers. Write down whether the reference retains disfluencies, filler words, repetitions, numbers as spoken or normalized digits, speaker labels, and punctuation. For punctuation, define how annotators should handle incomplete sentences and interruptions, and adjudicate disagreements rather than assuming that one mark is always uniquely correct.

6. Metrics that answer different questions

Question	Measure	Interpretation
Were words recognized?	Word error rate (WER)	Counts substitutions, deletions, and insertions relative to reference words. Publish normalization rules.
Were characters recognized?	Character error rate (CER)	Useful alongside WER; character normalization and whitespace policy affect results.
Was punctuation restored?	Per-mark precision, recall, and F1	Requires explicit word alignment, mark classes, and treatment of multiple or missing marks.
Are timestamps useful?	Boundary error against a timed reference	Report which boundaries were evaluated, units, and the median and distribution of errors.
Does the workflow complete?	Failure, timeout, and retry rates	Include failed and empty outputs; successful-run accuracy alone is incomplete.

WER is commonly expressed as (substitutions + deletions + insertions) / reference words. It is an error rate, not a percentage of words that are necessarily correct in a simple one-to-one sense, and it can exceed 100% when insertions are numerous. Report how case, punctuation, apostrophes, numerals, and whitespace were normalized before scoring. Avoid excluding difficult examples after seeing the result.

Punctuation scoring is more subtle. If the recognizer omits a word, simply comparing punctuation at the same word index can credit or penalize the wrong boundary. Align hypothesis words to reference words first and state how insertions, deletions, and unmatched boundaries are handled. Report per-class counts and precision, recall, and F1 as well as any aggregate. When a class has no reference examples or no predictions, document how its score is represented instead of silently substituting a value.

To evaluate timing, the reference must contain time annotations appropriate to the question: a segment-level reference cannot establish word-level timing accuracy. WhisperX presents one approach to word-level alignment [4], but an alignment tool's output is still a prediction and should be checked against a suitable reference. Also distinguish timestamps measured on VAD-processed audio from those mapped back to the original recording.

7. Reporting results without overstating them

Use paired comparisons when the same source audio is evaluated under different conditions. Report the number of source recordings, speakers where known, total duration, duration distribution, and the number of successful and failed runs per condition. When repeating calls, distinguish variation between recordings from variation between repeated calls on the same recording. Confidence intervals or another uncertainty estimate should respect the recording or speaker as the unit of sampling rather than treating correlated segments as independent.

For a segmentation study, compare matched transcript regions and include boundary-specific error analysis. For format tests, document the encoder and preprocessing path. For punctuation restoration, compare the recognizer's raw output with the post-processed output, and check

whether lexical content changed. A punctuation component that makes prose look fluent by rewriting or omitting spoken words may be unsuitable for a verbatim transcript even if the text reads well.

Finally, document limitations. A system evaluated on English read speech cannot automatically be characterized for other languages or spontaneous conversations. A small set of long recordings can reveal defects, but it cannot establish a general failure rate with high precision. If an upstream model is accessed through a hosted service and its internal version is unavailable, say so and record the date and exposed model identifier.

8. A practical investigation checklist

1. **Preserve the evidence.** With permission, retain the source audio, request settings, raw response, intermediate text, and final export. Remove credentials and personal information from any shared records.
2. **Locate the first missing mark.** Compare the raw response, stored data, interface, and export. This separates generation problems from later transformations.
3. **Control the inputs.** Fix language, model revision, VAD, decoding settings, and conversion parameters. Use the same source recording for each condition.
4. **Test alternatives.** Compare full input where supported, fixed-duration segmentation, and pause-aware boundaries. Repeat enough cases to distinguish a recurring pattern from a single run.
5. **Score and inspect.** Measure words and punctuation separately; inspect boundary windows, missing speech, duplicate text, and failure cases.
6. **Choose an intervention.** Fix the stage where the loss first appears. If a punctuation-restoration stage is used, verify that it preserves words and evaluate its output against human-reviewed text.

9. Conclusion

A useful transcription is not just a sequence of recognized words. Segmentation, punctuation, alignment, and export all affect whether a reader can trust and navigate it. The most productive way to investigate a defect is to keep the pipeline observable, compare matched conditions, and measure each output separately.

The protocol in this paper is intended as a starting point for that work. It does not establish that any particular model, boundary rule, or duration is best. Those conclusions require a representative dataset, verified references, and results reported with their limitations.

Transparency and scope

This article is a technical review and proposed evaluation protocol, not a report of new experimental results. It was prepared in connection with MP3ToText.io, a speech-to-text service; this relationship is disclosed so readers can assess the context. The article does not claim an independent evaluation of the service. The draft was prepared with AI assistance and requires human review, confirmation of authorship, and compliance with the publishing platform's disclosure requirements before publication.

References

[1] Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., and Sutskever, I. (2022). Robust Speech Recognition via Large-Scale Weak Supervision. *arXiv*. https://arxiv.org/abs/2212.04356

[2] Panayotov, V., Chen, G., Povey, D., and Khudanpur, S. (2015). LibriSpeech: An ASR corpus based on public domain audio books. *IEEE International Conference on Acoustics, Speech and Signal Processing*. https://doi.org/10.1109/ICASSP.2015.7178964

[3] Koenecke, A., et al. (2020). Racial disparities in automated speech recognition. *Proceedings of the National Academy of Sciences*. https://doi.org/10.1073/pnas.1915768117

[4] Bain, M., Huh, J., Han, T., and Zisserman, A. (2023). WhisperX: Time-Accurate Speech Transcription of Long-Form Audio. *arXiv*. https://arxiv.org/abs/2303.00747